

Analyst User Guide

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Attribute	Detail
Platform Scope	92 Nifty 100 companies · 50+ KPIs · FY 2010–2024
Modules	12 analytics modules · 120+ features
Interface	Streamlit dashboard (8 screens) + FastAPI (16 endpoints)
Reports	92 company tearsheets · 10 sector PDFs · 1 portfolio PDF
Test Coverage	171 automated tests · 0 critical failures

This guide covers: launching the dashboard, using the screener, reading peer comparisons, accessing PDF reports, and using the REST API. No programming knowledge required for daily use.

1. Getting Started

1.1 Launching the Dashboard

Open a terminal in the project folder and run:

```
streamlit run src/dashboard/app.py
```

The dashboard opens automatically at <http://localhost:8501> in your browser. Navigate between 8 screens using the left sidebar.

1.2 Launching the API

To start the REST API server:

```
uvicorn src.api.main:app --port 8000 --reload
```

Interactive API documentation is available at <http://localhost:8000/docs> — you can test all 16 endpoints directly from the browser without writing code.

1.3 Using Makefile Shortcuts

Command	What it does
make load	Rebuild the database from source Excel files
make ratios	Recompute all 50+ KPIs
make test	Run 171 automated tests
make report	Regenerate all 103 PDF reports
make dashboard	Start Streamlit on port 8501
make api	Start FastAPI on port 8000
make clean	Remove build artifacts (not the database)

2. Dashboard Guide — 8 Screens

2.1 Overview

Homepage with Nifty 100 summary KPIs, sector donut chart, health score distribution, and top-10 companies by health score. Use the KPI selector to compare sector medians.

2.2 Company Profile

Search any of 92 companies by ticker or name. Shows 12 KPI tiles, 5 chart tabs (Revenue/Profit, ROE/ROCE, Balance Sheet, Cash Flow, Valuation), and raw data tables. Download PDF tearsheet from this screen.

2.3 Screener

Filter companies using 10 sliders. Choose a preset (Quality Compounder, Value Pick, Growth Accelerator, etc.) or build a custom screen. Results export to CSV or Excel.

2.4 Peer Comparison

Select a peer group (11 available). View radar chart comparing selected company vs group average. Percentile heatmap shows green (top quartile) and red (bottom quartile).

2.5 Trend Analysis

Select any company + up to 3 metrics. Plots 10-year history with a year-on-year % change table and CAGR summary below.

2.6 Sector Analysis

Filter by sector or view all 92 companies. Bubble chart shows Market Cap vs ROE. Dual KPI bar charts compare sector medians.

2.7 Capital Allocation

Treemap of all 92 companies coloured by health score, grouped by capital pattern (Reinvestor, Leveraged Growth, Distress, etc.). Filter by pattern to see company list.

2.8 Annual Reports

Select any company and year range to access BSE annual report PDF links. Green badge = URL live, Red = missing.

3. Using the Investment Screener

The screener filters 92 companies across 10 metrics simultaneously. Results update live as you adjust sliders.

3.1 Pre-built Presets

Preset	Filters Applied	Use For
Quality Compounder	ROE>15%, D/E<1, FCF>0, RevCAGR5yr>10%	Long-term compounding
Value Pick	P/E<20, P/B<3, D/E<2, DivYield>1%	Undervalued stocks
Growth Accelerator	PATCAGR5yr>20%, RevCAGR5yr>15%, D/E<2	High-growth plays
Dividend Champion	DivYield>2%, Payout<80%, FCF>0	Income investors
Debt-Free Blue Chip	D/E=0, ROE>12%	Zero-debt fortress
Turnaround Watch	RevCAGR3yr>10%, FCF positive	Recovery plays

3.2 Composite Score

Every company receives a Composite Score (0–100) calculated as: Profitability 35% (ROE+ROCE+NPM) + Cash Quality 30% (FCF CAGR+CFO/PAT) + Growth 20% (Rev CAGR+PAT CAGR) + Leverage 15% (D/E+ICR). Results are always sorted by this score descending.

3.3 Customising Filters

To create a custom screen, open config/screener_config.yaml and add a new entry under 'presets:'. No code changes are needed. The YAML file is self-documented with all available filter keys.

4. KPI Quick Reference

KPI	Formula	Benchmark	Flag If
ROE	$\text{Net Profit} / \text{Equity} \times 100$	>15% = good	<8% = weak
ROCE	$\text{EBIT} / \text{Capital Employed} \times 100$	>15% = good	N/A for banks
NPM	$\text{Net Profit} / \text{Sales} \times 100$	>10% = good	<5% = thin
D/E	$\text{Borrowings} / \text{Total Equity}$	<1.0 = healthy	>2 non-fin
ICR	$(\text{Op Profit} + \text{Other Inc}) / \text{Interest}$	>3× = safe	<1.5 = danger
FCF	$\text{CFO} + \text{CFI}$	>0 = generating	<0 3yr = flag
RevCAGR 5yr	Revenue CAGR over 5 years	>10% = healthy	< 5% = slow
PATCAGR 5yr	Net Profit CAGR over 5 years	>15% = healthy	Turnaround=None
Health Score	Composite 0–100	70+ = Excellent	< 35 = Weak
CFO/PAT	$\text{Operating CF} / \text{Net Profit}$	>1.0 = quality	<0.5 = accrual

5. REST API Usage

The API runs at <http://localhost:8000>. All responses are JSON. Interactive docs at </docs>. No authentication required (internal network only).

Endpoint	Returns
GET /api/v1/health	Server status + DB row counts
GET /api/v1/companies	All 92 companies (filter: ?sector=IT)
GET /api/v1/companies/TCS	Full profile + latest KPIs
GET /api/v1/companies/TCS/ratios	All 50+ KPIs for all years
GET /api/v1/companies/TCS/tearsheet	Download PDF tearsheet
GET /api/v1/screener?min_roe=15&max_de=1	Filtered company list
GET /api/v1/sectors	11 sectors with median KPIs
GET /api/v1/peers/IT Services	Peer group with percentile ranks
GET /api/v1/portfolio/stats	P10–P90 for 10 KPIs
GET /api/v1/market-cap/TCS	Historical P/E, P/B, EV/EBITDA

5.1 Example API Calls

```
# Get all IT sector companies curl
http://localhost:8000/api/v1/sectors/Information%20Technology/companies

# Screen for quality stocks curl
"http://localhost:8000/api/v1/screener?min_roe=15&max_de=1&min_fcf=0"

# Download TCS tearsheet PDF curl -O
http://localhost:8000/api/v1/companies/TCS/tearsheet
```

6. PDF Reports

6.1 Company Tearsheets (92 PDFs)

Location: reports/tearsheets/ | Format: _tearsheet.pdf | 2 pages per company. Page 1: KPI tiles + revenue/profit bar + ROE/ROCE trend. Page 2: Balance sheet composition + cash flow + pros/cons.

6.2 Sector Reports (10 PDFs)

Location: reports/sector/ | Format: _report_YYYYMMDD.pdf. Each report includes: sector median KPIs, best/worst 3 companies highlighted, full ranked company table with colour-coded health bands.

6.3 Portfolio Summary

Location: reports/portfolio/ | Format: portfolio_summary_YYYYMMDD.pdf. Cover page with universe statistics. Then all 92 companies, 4 per page, with trend arrows (↑ ↓ →) showing 3-year direction for ROE, NPM, FCF, Rev CAGR.

6.4 Regenerating Reports

To regenerate all 103 PDFs:

```
make report
```

Or regenerate individually: python src/reports/tearsheet.py TCS

7. Data Quality Notes for Analysts

Item	Note
SBIN	No balance sheet data in source — excluded from KPI computation
Bank D/E ratios	Banks show D/E > 5× — structurally normal, NOT a red flag
ROE > 100%	BEL, HAL, INDIGO have near-zero equity — ROE capped at 200%
CAGR = None	Turnaround companies (loss → profit) show None, not a missing value
Market cap data	2019–2024 valuation data is simulated for learning purposes
Stock prices	Monthly OHLCV is simulated — do not use for trading decisions
Distress flags	Some bank/NBFC distress flags are artifacts of lending business model
8 missing companies	WIPRO, ULTRACEMCO etc. were in P&L but not in companies master table
pros_cons source	641 entries are auto-generated via KPI rules — confidence shown per entry

All simulated datasets (market_cap, stock_prices) are clearly labelled in both the Streamlit dashboard and PDF reports.

8. Troubleshooting

Problem	Solution
Dashboard won't start	Run: <code>pip install streamlit plotly openpyxl pyyaml</code>
Database not found	Run: <code>make load</code> (or: <code>python src/etl/loader.py</code>)
No KPI data showing	Run: <code>make ratios</code> (<code>computed_ratios</code> table is empty)
API returns 503	Database may be locked — close other connections
Tearsheet PDF missing	Run: <code>python src/reports/tearsheet.py <TICKER></code>
Port 8501 already in use	Kill existing Streamlit: <code>pkill -f streamlit</code>
Port 8000 already in use	Change port in <code>.env</code> : <code>PORT=8001</code>
Tests failing after data update	Run: <code>make load</code> then <code>make ratios</code> before <code>make test</code>
SBIN missing from screener	Known — SBIN has no balance sheet data in source files